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MILITARY SPECIFICATION

HARDWARE, INDIVIDUAL LOAD CARRYING EQUIPMENT: AND HARDWARE, MISCELLANEOUS

This specification is approved for use by the U.S. Army Natick Research, Development, and Engineering Center, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document covers hardware for individual load carrying equipment.

1.2 Classification. Hardware shall be of the following types and classes as specified:

Type I	- Fastener, Belt, Pistol, Female, Round Stud
Type II	- Fastener, Belt, Pistol, Male, Round Stud
Type III	- Fastener, Pouch, Ammunition, Male
Type IV	- Fastener, Pouch, Ammunition, Female
Type V	- Fastening, End, 1 inch, with Loop and Clip; Right Hand
Type VI	- Fastening, End, 1 inch, with Loop and Clip; Left Hand
Type VII	- Fastening, End, 1 inch, with Clip
Type VIII	- Keeper, Clamp, 5/8 inch

Class A	- Brass
Class B	- Steel

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: U.S. Army Natick Research, Development, and Engineering Center, Natick, MA 01760-5014 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 5340

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Type IX - Keeper, Clamp, 1 inch, with Loop

Class A - Brass
Style 1 - For use with cotton webbing
Style 2 - For use with nylon webbing

Class B - Steel
Style 1 - For use with cotton webbing
Style 2 - For use with nylon webbing

Type X - Keeper, with Slide
Type XI - Keeper, Sliding, 2-5/16 inch

Class A - Brass
Class B - Steel

Type XII - Buckle, Tongueless, Double Bar, 5/8 inch, with Lip

Class A - Brass
Class B - Steel

Type XIII - Fastener, Belt, Male and Female, Aluminum
Type XIV - Buckle, 1 inch Non-Slip, Quick Release, Pack Suspenders
Type XV - Buckle, Clinch, 2-1/4 inch Adjustment, Pistol Belt

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

FEDERAL

QQ-A-250/11 - Aluminum Alloy 6061, Plate and Sheet
QQ-B-613 - Brass, Leaded and Nonleaded: Flat Products (Plate, Bar, Sheet, and Strip)
QQ-C-390 - Copper Alloy Castings (Including Cast Bar)
QQ-S-700 - Steel Sheet and Strip, Medium and High Carbon
QQ-W-461 - Wire, Steel, Carbon, (Round, Bare, and Coated)
TT-E-529 - Enamel, Alkyd, Semigloss
TT-P-1757 - Primer Coating, Zinc Chromate, Low-Moisture-Sensitivity

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PPP-B-566 - Boxes, Folding, Paperboard
PPP-B-621 - Boxes, Wood, Nailed and Lock-Corner
PPP-B-636 - Boxes, Shipping, Fiberboard
PPP-B-640 - Boxes, Fiberboard, Corrugated, Triple-Wall
PPP-B-665 - Boxes: Paperboard, Metal Edged and Components
PPP-B-676 - Boxes, Setup

MILITARY

MIL-F-495 - Finish, Chemical, Black, for Copper Alloys
MIL-C-5541 - Chemical Conversion Coatings on Aluminum and Aluminum Alloys
MIL-H-6088 - Heat Treatment of Aluminum Alloys
MIL-A-8625 - Anodic Coatings for Aluminum and Aluminum Alloys
MIL-L-10547 - Liners, Case, and Sheet, Overwrap; Water-Vaporproof or Waterproof, Flexible
MIL-T-12879 - Treatment, Chemical, Prepaint and Corrosion Inhibitive, for Zinc Surfaces
MIL-C-13924 - Coating, Oxide, Black for Ferrous Metals
MIL-L-35078 - Loads, Unit: Preparation of Semiperishable Subsistence Items; Clothing, Personal Equipment and Equipage, General Specification For

STANDARDS

MILITARY

MIL-STD-105 - Sampling for Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage
MIL-STD-147 - Palletized Unit Loads

(Copies of specifications, standards, and handbooks required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

DRAWINGS

U.S. ARMY NATICK RESEARCH, DEVELOPMENT, AND ENGINEERING CENTER

4-1-137 - Hardware, Individual Load Carrying Equipment; Fastener, Belt, Pistol, Fastener, Pouch; AMMO; Fastening End, 1 inch, With and Without Loop; Clip; Assembly and Details

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4-1-138 - Hardware, Individual Load Carrying Equipment; Keeper, Clamp, 1 inch, with Loop and Keeper, Clamp, 5/8 inch, Assembly and Details
4-1-139 - Hardware, Individual Load Carrying Equipment; Keeper, with Slide, Assembly, Keeper, Sliding, 2-5/16 inch, and Buckles, Tongueless, Double Bar, 5/8 inch with Lip
4-1-231 - 1 inch Non-Slip, Quick Release Buckle Assembly with Spring
4-1-443 - Hardware, Individual Load Carrying Equipment: Fastener, Belt, Male and Female
4-1-453 - Buckle, Clinch, Adjustment, 2-1/4, Assembly

(Copies of drawings, publications, and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS shall be the issues of the nongovernment documents which are current on the date of the solicitation.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

A 109 - Steel, Carbon, Cold-Rolled Strip
A 366 - Steel, Carbon, Cold-Rolled Sheet, Commercial Quality
A 569 - Steel, Carbon (0.15 Maximum, Percent), Hot-Rolled Sheet and Strip, Commercial Quality
B 183 - Preparation for Low-Carbon Steel for Electroplating
B 242 - Preparation of High-Carbon Steel for Electroplating
B 633 - Electrodeposited Coating of Zinc on Iron and Steel
D 1005 - Measurement of Dry-Film Thickness of Organic Coatings using Micrometers
D 3951 - Standard Practice for Commercial Packaging
E 18 - Rockwell Hardness and Rockwell Superficial Hardness of Metallic Materials

(Copies should be obtained from the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103-1187.)

(Nongovernment standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

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2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection (see 4.3, 6.2, and 6.3).

3.2 Materials and components. Materials and components shall be as specified herein and on the applicable drawings. Materials not definitely specified shall be of the quality normally used by the manufacturer in his standard commercial fastener, keeper, or buckle provided the completed item complies with all provisions of this document (see 6.4).

3.2.1 Brass.

3.2.1.1 Castings. Brass castings shall conform to copper alloys No. C 857, C 854, or C 836 of QQ-C-390.

3.2.1.2 Sheet or strip. The brass sheet or strip shall conform to alloy 260 or 268, quarter-hard, of QQ-B-613.

3.2.2 Steel.

3.2.2.1 Steel wire. The steel wire shall conform to steel number 1006-1010, finish 1, hard temper of QQ-W-461.

3.2.2.2 Sheet or strip. The sheet shall conform to HRCQ of ASTM A 569 or CRCQ of ASTM A 366. The strip shall be hot rolled commercial quality of ASTM A 569 or cold rolled No. 2 or No. 3 temper of ASTM A 109.

3.2.2.3 Sheet or strip (medium carbon). The sheet or strip shall conform to steel numbers 1025-1050 of QQ-S-700.

3.2.2.4 Aluminum alloy. Aluminum alloy shall conform to 6061 of QQ-A-250/11.

3.3 Construction. The hardware shall be constructed in accordance with Drawings 4-1-137, 4-1-138, 4-1-139, 4-1-231, 4-1-443, and 4-1-453 as applicable, and as specified herein.

3.3.1 Types I and II hardware. The types I and II hardware shall be cast from alloy specified in 3.2.1.1.

3.3.2 Types III and IV hardware. The loops for the types III and IV hardware shall be formed of steel wire as specified in 3.2.2.1. The clips shall be stamped from steel as specified in 3.2.2.2.

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3.3.3 Types V, VI and VII hardware. The loops for the types V, VI, and VII hardware shall be stamped from steel as specified in 3.2.2.3. After forming, the loops shall be heat treated to the hardness specified on Drawing 4-1-137. Testing for hardness shall be as specified in 4.4.2.2. The clips for the types V, VI, and VII hardware shall be stamped from steel as specified in 3.2.2.2, or from alloy as specified in 3.2.1.2.

3.3.4 Types VIII and IX hardware. The body and latch of the type VIII and type IX hardware shall be stamped from steel as specified in 3.2.2.2 or alloy as specified in 3.2.1.2 (see 1.2).

3.3.5 Type X hardware. The keeper, body, and slide for the type X hardware shall be stamped from steel as specified in 3.2.2.3 and heat treated to the hardness specified on Drawing 4-1-139. Testing for hardness shall be as specified in 4.4.2.2. The hardware shall be assembled so that the force to open the slide from the closed position shall be from 2 to 8 pounds when tested as specified in 4.4.5.

3.3.6 Types XI and XII hardware. The types XI and XII hardware shall be stamped from steel as specified in 3.2.2.2 or alloy as specified in 3.2.1.2 (see 1.2).

3.3.7 Type XIII hardware. The type XIII hardware shall be stamped from aluminum as specified in 3.2.2.4 and after forming heat treated to temper T-4 in accordance with MIL-H-6088.

3.3.8 Type XIV, hardware. The type XIV hardware shall be stamped and cast from material specified on Drawing 4-1-231.

3.3.9 Type XV, hardware. The type XV hardware shall be stamped from material specified on Drawing 4-1-453.

3.4 Finish. Any welding or brazing required shall be completed prior to the finish operation. Components of the type X hardware shall be finished prior to assembly.

3.4.1 Brass hardware. The brass hardware shall be given a black chemical finish conforming to MIL-F-495.

3.4.2 Steel hardware.

3.4.2.1 Steel, carbon. The carbon steel hardware shall be given a phosphate treated zinc plate conforming to type IV (with a minimum thickness of 13 micrometers) of ASTM B 633 followed by bake enameling. The bake enamel shall conform to type II of TT-E-529. The color shall be black.

3.4.2.1.1 Steel, carbon alternate finish. As an alternate to the phosphate treated zinc plate specified in 3.4.2.1, the carbon steel may be base plated using an electrodeposited brass coating as follows:

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a. The base metal shall be cleaned and prepared in accordance with ASTM B 183 or ASTM B 242, as applicable.

b. The hardware shall then be brass plated to a minimum thickness of 0.0004 inches by electrodeposition of brass alloy number 260 of QQ-B-613.

c. The hardware shall then be bake enameled as specified in 3.4.2.1.

3.4.2.2 Steel, corrosion resistant. The finish for CRES (stainless) components of types XIV and XV shall be a black oxide coating in accordance with MIL-C-13924.

3.4.3 Aluminum hardware. All surfaces of the aluminum hardware shall be given a chemical treatment conforming to class 1a of MIL-C-5541. After the chemical treatment, all treated surfaces shall be coated with primer conforming to composition G or L, as applicable, and color Y of TT-P-1757, followed by bake enameling. The enamel shall conform to type II of TT-E-529. The color shall be black.

3.4.3.1 Optional finish, aluminum hardware. As an option, aluminum hardware components may be anodized in accordance with type II, class 2 of MIL-A-8625. The color shall be black.

3.4.4 Zinc alloy. The alternate zinc alloy allowed on Drawing 4-1-231 for the tongue component of type XIV shall be finished as follows: Zinc surfaces shall be prepared for painting in accordance with type I, class 1 requirements of MIL-T-12879. Primer and black enamel coating shall be as specified in 3.4.3 for aluminum surfaces.

3.4.5 Epoxy powder coating. As an alternate to the finishes specified in 3.4.1 through 3.4.4, the surfaces may be prepared and coated as follows:

a. The hardware shall be prepared for coating by mechanical means. The surfaces shall be thoroughly and uniformly textured by sand blasting with 80-grit aluminum oxide. Prepared surfaces shall be kept clean during subsequent handling. Handling is to be by clean-gloved hands or by clean implements. The surfaces shall be retained in a clean environment. The surfaces shall be epoxy coated no more than 8 hours after sand blasting.

b. The finish coating shall be black electrodeposited powder coating conforming to H. B. Fuller Company, Kativo Powder No. KEP43286-20 or equal.

c. The electrodeposited epoxy powder coating shall be in accordance with commercial practice.

d. The thickness of the baked coating shall be a minimum of 0.003 inch $\pm$ 0.001 inch when examined as specified in 4.4.4.1.

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e. The final coating shall be smooth and uniform in appearance, free from wrinkles and streaks.

3.5 Marking for identification. All hardware shall be marked with the manufacturer's identification. The identification marks shall be incorporated in such a manner that the surfaces opposite those markings shall show no signs of penetration or unevenness or cause distortion in any part of the hardware.

3.6 Workmanship. The finished hardware shall be free from dirt and other harmful extraneous material. The coating shall level out to a smooth and uniform film without orange peel, wrinkles, drops, streaks or areas of no film.

3.6.1 Stampings. The stampings shall be clean, smooth, and free from cracks, burrs, drag, step, and rough edges.

3.6.2 Castings. The castings shall be smooth and free from blowholes, porosity, cracks, and shrinkage.

3.6.3 Welds. Where welding is required, the welds shall be sound, smooth, and free from porosity, holes, fissures, and burns or reduced areas.

3.6.4 Brazing. When brazing is required, brazing shall be complete, smooth, and uniform. All flux deposits shall be removed.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this document where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this document shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirement in the document shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

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a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).

4.3 First article inspection. When a first article is required (see 6.2), it shall be examined for the defects specified in table II and tested for the characteristics specified in 4.5.1. The presence of any defect or failure of any test shall be cause for rejection of the first article.

4.4 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

4.4.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced documents unless otherwise excluded, amended, modified, or qualified in this document or applicable purchase document.

4.4.2 In-process inspection.

4.4.2.1 Process examination. Examination shall be made to determine that the requirements specified in table I are followed. Whenever nonconformance is noted, correction shall be made to the item affected and process.

TABLE I. In-process examinations

Requirement operation	Paragraph
Heat treatment	3.3.7
Phosphate treated zinc plate	3.4.2.1
Chemical treatment	3.4.3
Primer coat	3.4.3
Finish prior to assembly (type X)	3.4
Zinc alloy preparation and primer coat	3.4.4

4.4.2.2 In-process testing. Testing of the fabricated components of types V, VI, VII, and X hardware shall be performed in accordance with table II for the characteristics shown therein. The sample unit for types V, VI, and VII hardware shall be one loop of each type. The sample unit for type X hardware shall be one of each, keeper, body, and slide. Testing of the hardware shall be accomplished prior to the finish and assembly.

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TABLE II. In-process tests

Component and unit of product	Characteristics	Requirement reference	Test method	Requirement defects applicable to individual unit	Number of determinations per sample unit	Results reported as
Types V, VI, and VII hardware (loops)	Hardness Rockwell C36-44	Drawing 4-1-137	ASTM E 18	X	3	Average of 3 determinations to nearest Rockwell hardness number
Type X hardware (keeper)	Hardness Rockwell 15N-74-83	Drawing 4-1-139	ASTM E 18	X	3	Average of 3 determinations to nearest Rockwell hardness number
(body)	Hardness Rockwell C35-45	Drawing 4-1-139	ASTM E 18	X	3	Average of 3 determinations to nearest Rockwell hardness number
(slide)	Hardness Rockwell C35-45	Drawing 4-1-139	ASTM E 18	X	3	Average of 3 determinations to nearest Rockwell harness number

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4.4.3 End item visual examination. The end items shall be examined for the defects listed in table III. The lot size shall be expressed in units of hardware. The sample unit shall be one type and class of hardware offered for inspection at one time. The inspection level shall be II and the AQL, expressed in terms of defects per hundred units, shall be 2.5 for major defects and 6.5 for total (major and minor combined) defects.

TABLE III. End item visual defects

Examine	Defect	Classification	
		Major	Minor
Finish	Not black chemical anodized or enamel finished as applicable	101	
	Area of no film	102	
	Finish not adherent, orange peel		201
	Wrinkles, drops, or streaks		202
Design	Component not formed as specified in applicable drawings, e.g., fastening end, 1 inch, with loop, does not conform to design shown, or clip for fastener, pouch, ammunition, female is not formed to shape as specified in Drawing 4-1-137	103	
Construction and workmanship	Not properly assembled	104	
	Any part missing	105	
	Fractured, split, bent, dented, sprung, or malformed	106	
	Any sharp burr, sliver, or sharp edge	107	
	Holes in hardware missing where required	108	
	Holes in hardware misplaced or out of alignment, affecting assembly	109	
	Holes in hardware too large or too small	110	
	Keeper clamp latch cannot be operated by thumb without the aid of a tool		203

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TABLE III. End item visual defects - Continued

Examine	Defect	Classification	
		Major	Minor
Construction and workmanship - continued	Keeper clamp latch requires no force or slight force to operate	111	
	NOTE: Opening of latch of keeper, clamp becomes easier with repeated operations. Tight fit is preferred, provided it can be operated by thumb.		
	Slide cannot be operated	112	
	Slide is too loose, (i.e., will operate without moderate push or pull)		204
	Slide tip contacts any portion of the slot in the body end while activating the slide	113	
Welding and brazing	Missing, fissures, burns, holes, or reduced areas	114	
	Incomplete, porous		205
	Not smooth and uniform, or scale or flux deposits not removed	115	
	Loops not welded or brazed to clips where required	116	
Identification markings	Missing, incomplete, illegible, or incorrect		206
	Not stamped or etched on hardware as specified		207
	Indication of penetration, unevenness, or distortion marking on hardware		208

4.4.4 End item dimensional examination. The end items shall be examined for the dimensions designated with an asterisk on specified drawings. Any designated dimension not within the tolerances specified shall be classified as a defect. The lot size shall be expressed in units of hardware. The sample unit shall be one type and class of hardware. The inspection level shall be S-2 and the AQL, expressed in terms of defects per hundred units, shall be 4.0.

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4.4.4.1 Epoxy coating thickness examination (when the optional epoxy powder coating is used). The end items shall be examined for thickness of the epoxy coating in accordance with ASTM D 1005. Three observations shall be made per sample unit. Any epoxy coating thickness less than specified in 3.4.5 shall be classified as a defect. The lot size shall be expressed in units of hardware of one type and class only. The sample unit shall be one assembled and finished piece of hardware. The inspection level shall be S-2 and the AQL, expressed in terms of defects per hundred units, shall be 4.0.

4.4.5 End item testing. The end items, type X only, shall be tested for the characteristics specified in table III. The lot size shall be expressed in units of hardware of one type and class only. The sample unit shall be one completely fabricated unit of hardware. For all testing, the inspection level shall be S-1 and the AQL, expressed in terms of defects per hundred units for each characteristic, shall be 6.5.

TABLE III. End item tests

Characteristic	Requirement paragraph	Test method	Number of determinations per sample unit	Results reported as
Force to start slide (applicable to type X hardware)	3.3.5	4.5.1	1 (after the first cycle)	Nearest 1.0 ounce

4.4.6 Packaging examination. The fully packaged end items shall be examined for the defects listed below. The lot size shall be expressed in units of shipping containers. The sample unit shall be one shipping container fully packaged. The inspection level shall be S-2 and the AQL, expressed in terms of defects per hundred units, shall be 2.5.

Examine	Defect
Marking (exterior and interior)	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application
Materials	Any component missing, damaged, or not as specified

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Examine	Defect
Workmanship	Inadequate application of components, such as incomplete sealing or closure of flap, improper taping, loose strapping, or inadequate stapling Bulged or distorted container
Content	Number per container is more or less than required

4.4.7 Palletization examination. The fully packaged and palletized end items shall be examined for the defects listed below. The lot size shall be expressed in units of palletized unit loads. The sample unit shall be one palletized unit load, fully packaged. The inspection level shall be S-1 and the AQL, expressed in terms of defects per hundred units, shall be 6.5.

Examine	Defect
Finished dimensions	Length, width, or height exceeds specified maximum requirement
Palletization	Pallet pattern not as specified Interlocking of loads not as specified Load not bonded as specified
Weight	Exceeds maximum load limits
Marking	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application

4.5 Methods of inspection.

4.5.1 Force test. The force test may be applied by means of a spring balance or any force measuring device in accordance with the manufacturer's standard practice.

5. PACKAGING

5.1 Preservation. Preservation shall be level A or Commercial as specified (see 6.2).

5.1.1 Level A. Unless otherwise specified (see 6.2), hardware of one type and class only shall be unit packed in the quantities specified in table IV in a snug-fitting fiberboard box conforming to style RSC, type CF or SF, class domestic, grade 200 of PPP-B-636. When quantities per unit pack other than those in table IV are specified, hardware shall be unit packed in boxes conforming to PPP-B-566, PPP-B-665, PPP-B-676 or PPP-B-636. Each box shall be agitated from time to time to assure a compact and well-filled box. Closure

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shall be in accordance with the appendix of the applicable box document. When specified (see 6.2), boxes conforming to style RSC, type CF or SF, class weather resistant, grade W5c or W6c of PPP-B-636, closed and waterproofed in accordance with the appendix of PPP-B-636, shall be used.

TABLE IV. Hardware packaging quantities

Type and class of hardware	Quantity per unit pack	Quantity per shipping container
Type I	350	2100
Type II	300	1800
Type 1II	1300	7800
Type IV	1000	6000
Type V	600	3600
Type VI	600	3600
Type VII	800	4800
Type VIII		
Class A	700	4200
Class B	700	4200
Type IX		
Class A	400	2400
Style 1		
Style 2		
Class B	400	2400
Style 1		
Style 2		
Type X	400	2400
Type XI		
Class A	800	4800
Class B	800	4800
Type XII		
Class A	1750	10,500
Class B	2000	12,000

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TABLE IV. Hardware packaging quantities - Continued

Type and class of hardware	Quantity per unit pack	Quantity per shipping container
Type XIII	100	1000
Type XIV	300	1800
Type XV	300	1800

5.1.2 Commercial. Hardware of one type and class only, as specified in table IV, shall be preserved in accordance with ASTM D 3951.

5.2 Packing. Packing shall be level A, B, or Commercial as specified (see 6.2).

5.2.1 Level A packing. Unless otherwise specified (see 6.2), hardware of one type and class only, preserved as specified in 5.1, shall be packed in the quantities specified in table IV in snug-fitting fiberboard shipping containers conforming to style FTC, V2s of PPP-B-636, except that weight limitation shall be waived and the inside dimensions of the container (length, width and depth added) shall not exceed 48 inches; or a nailed wood shipping container conforming to class 2, style 2 of PPP-B-621. When quantities per shipping container other than those in table IV are specified, hardware shall be packed in snug-fitting shipping containers conforming to style RSC, class weather-resistant grade V2s of PPP-B-636, or class 2 of PPP-B-621, in accordance with the weight limitation of the applicable box document. Each fiberboard shipping container shall be waterproofed, closed, and reinforced with flat strapping or tape banding in accordance with PPP-B-636. Each wood shipping container shall be provided with the type I or II, grade C case liner conforming to MIL-L-10547 and closed and strapped in accordance with PPP-B-621. When hardware is packaged in boxes conforming to W5c or W6s as specified in 5.1.1, waterproofing will not be required. Shipping containers shall be arranged in unit loads in accordance with MIL-L-35078 for the type and class specified (see 6.2). When unit loads are strapped, strapping shall be limited to nonmetallic strapping, except type II, class F loads.

5.2.2 Level B packing. Unless otherwise specified (see 6.2), hardware of one type and class only, preserved as specified in 5.1, shall be packed in the quantities specified in table IV in a snug-fitting fiberboard shipping container conforming to style FTC, type CF, class domestic, variety DW, grade 350 of PPP-B-636 or class 1 of PPP-B-640, except that weight limitation shall

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be waived and the inside dimensions of the container (length, width, and depth added) shall not exceed 48 inches; or a nailed wood shipping container conforming to class 1, style 2 of PPP-B-621. When quantities per shipping container other than those in table IV are specified, hardware shall be packed in snug-fitting shipping containers conforming to style RSC, class domestic of PPP-B-636, class 1 of PPP-B-621, or class 1 of PPP-B-640 in accordance with the weight limitation of the document. Closure shall be in accordance with the appendix of the container document. When specified (see 6.2), the shipping container shall be V3c, V3s, or V4s, fabricated and closed in accordance with PPP-B-636.

5.2.3 Commercial packing. Hardware, preserved as specified in 5.1.2, shall be packed according to ASTM D 3951.

5.3 Palletization. When specified (see 6.2), hardware, packed as specified in 5.2.2 and 5.2.3, shall be palletized on a 4-way entry pallet in accordance with load type Ia of MIL-STD-147. Pallet type shall be type I (4-way entry), type IV or type V in accordance with MIL-STD-147. Each prepared load shall be bonded with primary and secondary straps in accordance with bonding means K and L or film bonding means O or P. Pallet pattern shall be in accordance with the appendix of MIL-STD-147. Interlocking of loads shall be effected by reversing the pattern of each course.

5.4 Marking. In addition to any special marking required by the contract or purchase order, unit packs, shipping containers, and palletized unit loads shall be marked in accordance with MIL-STD-129 or ASTM D 3951, as applicable.

6. NOTES

6.1 Intended use. This document covers hardware intended to be used on individual load carrying equipment.

6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this document.
b. Type, class, and style (when applicable) of hardware required (see 1.2).
c. When a first article is required (see 3.1, 4.3, and 6.3).
d. Selection of applicable levels of preservation and packing (see 5.1 and 5.2).
e. If quantity of hardware per package and shipping container differ from that specified in table IV (see 5.1.1, 5.2.1, and 5.2.2).
f. When weather-resistant grade fiberboard shipping containers are required for level A packing (see 5.1.1).
g. Type and class of unit load required (see 5.2.1).
h. When palletization is required (see 5.3).

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6.3 First article. When a first article is required, it shall be inspected and approved under the appropriate provisions of FAR 52.209. The first article should be a preproduction sample. The contracting officer should specify the appropriate type of first article and the number of units to be furnished. The contracting officer should include specific instructions in all acquisition instruments regarding arrangements for selection, inspection, and approval of the first article.

6.4 Recycled material. It is encouraged that recycled material be used when practical as long as it meets the requirements of this document (see 3.2).

6.5 Subject term (key word) listing.

Equipment, individual load carrying
Fastener, belt
Fastener, pouch
Keeper, clamp

6.6 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodian:

Army - GL

Review activities:

Army - AR, AV, EA, MI, ME
DLA - IS

Preparing activity:

Army - GL

Project No. 5340-1845

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-H-9890E(GL)	Hardware, Individual Load Carrying Equipment: And Hardware, Miscellaneous

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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